Can LLMs Quantify Market Reactions to Financial News and Regulation?

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Introduction

Motivation

Financial regulators such as the SEC, FDIC, and the Federal Reserve closely monitor market stability, yet it remains difficult to measure how quickly and intensely financial markets react to regulatory announcements and news events.

We ask 3 Questions

1. Can sentiment scores derived from Large Language Models (LLMs) applied to financial news predict short-term volatility in regional bank stocks?
2. Do LLM-based sentiment indicators provide earlier or stronger signals of market instability than traditional market metrics such as historical volatility or trading volume?
3. Are market reactions to financial news and regulation geographically concentrated, affecting certain regions more than others?

Figure out how AI-based text analysis could support financial stability monitoring and regulatory decision-making

Data

RAW DATA

1. Financial News & Regulatory Text Data

   Daily financial headlines from Kaggle and other news partners, with timestamps and tickers.

   Used to train a light sentiment classifier (positive / negative / neutral) and to build yearly indices of regulatory-news sentiment.

2. Banking fundamentals (FDIC) Data

   FDIC state-level "Summary of Results" for commercial banks – financial.

   State–year totals for assets, net income, number of banks, net interest margin (NIM), and related balance-sheet items.

3. Geographic Data for Banks

   U.S. Census TIGER/Line shapefiles for U.S. states (cb_2024_us_state_20m).

   Provides state boundaries used to draw choropleth maps of banking stress.

Data

FDIC *banking fundamentals data : ROA, ROE, Deposits etc*

negative regulatory/banking regulation news ➞NLP➞sentiment index

1. Merge FDIC ***state-year banking fundamentals*** with ***yearly regulatory-news sentiment***.

❶ **ΔROA (Profitability Change)**
Change in return on assets for state s from year t to t+1

$$\Delta ROA_{s,t} = ROA_{s,t+1} - ROA_{s,t}$$

❷ **bad_year (Stress Indicator)**
Binary indicator equal to 1 if ΔROA is unusually low
If ΔROA very low, bad_year = 1

❸ **severity (Magnitude of Deterioration)**
Continuous measure capturing how severe the profitability decline is (HOW bad it was)

❹ StressScore (Composite Stress Index)
StressScore = Probability of Distress × Severity of Distress

$$StressScore = P(bad_year) \times E(severity)$$

2.Visualization Outputs

Static scatter: regulatory-news negativity vs. ΔROA across state–years.

Static map: state-level StressScore

Method - Two-Stage Prediction + Stress Score

Do Sentiment Variables Enhance Bank Stress Prediction?

Model 1: Bank Stress ~ FDIC fundamentals

Model 2: Bank Stress ~ FDIC fundamentals + News Sentiment

Model	AUC (Bad Year Prediction)	R^2 (Severity Prediction)
FDIC Fundamentals Only	0.636	-0.164
FDIC + News Sentiment	**0.646**	**-0.126**

AUC → The model's ability to distinguish between "stressed" and "normal" states.

Higher AUC = better classification

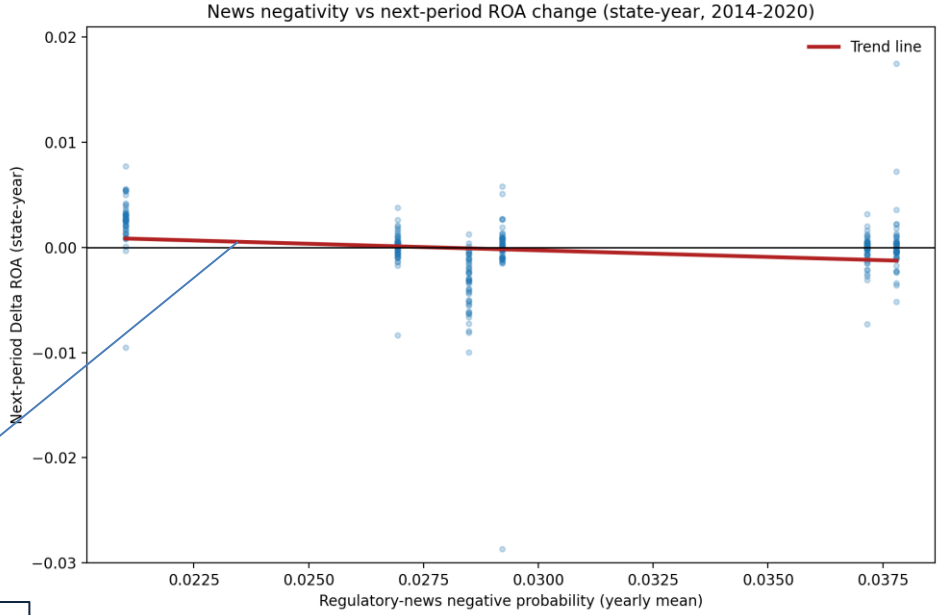
R^2 → The model's ability to explain the extent of bank profitability deterioration.

Higher R^2 = better regression fit

Adding regulatory-news sentiment slightly improves prediction performance, suggesting that news tone contains additional information about banking stress.

Result 1: Is regulatory news sentiment correlated with changes in bank profitability?

More negative regulatory news sentiment
↓
Predicts / Is associated with
↓
Deterioration in next-period bank profitability (ROA declines)
↓
If this relationship holds
↓
News sentiment may serve as an early warning signal of banking stress

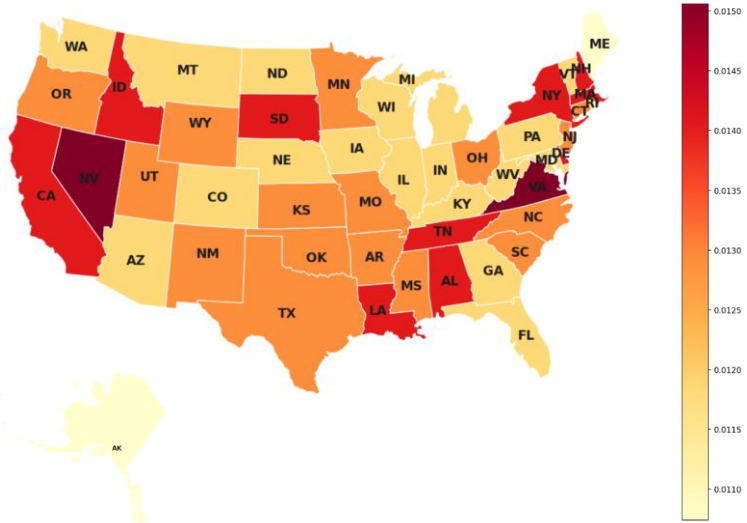


As regulatory news turns more negative

Changes in bank profitability in the following period show a declining trend.

More negative regulatory news sentiment is associated with a higher likelihood of deterioration in next-period bank profitability. This reflects a **directional relationship**, rather than a **strong linear relationship**.

Result 2: Geographic Distribution of Banking Stress



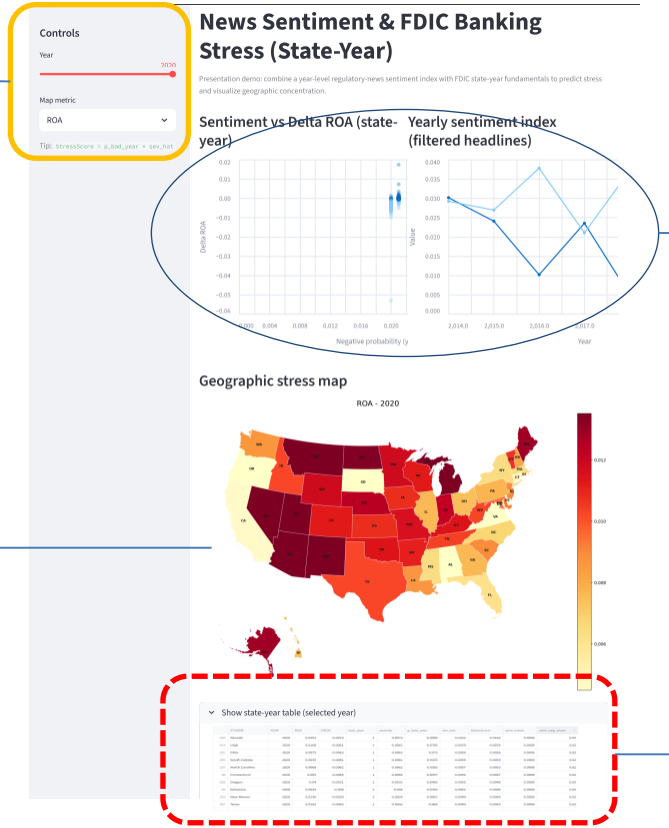
Top Stress States		
State	Bad-Year Share	Worst ΔROA
Nevada	**0.71**	-0.0287
Virginia	**0.71**	-0.0064
Delaware	0.43	-0.0080
South Dakota	0.43	-0.0081
Alabama	0.43	-0.0100

Most Sentiment-Sensitive States	
State	ΔROA if negative news +10pp
Virginia	**-0.0439**
Tennessee	-0.0376
South Dakota	-0.0323
South Carolina	-0.0304
Delaware	-0.0279

- Nevada and Virginia show the highest frequency of stress years in the sample.
- The lagged impact of negative regulatory news on profitability is strongest in VA, TN, SD, and SC, highlighting these states' heightened sensitivity to sentiment shocks.

Live Monitoring Dashboard & Key Takeaways

❶ Users can select the **year** and **map** metrics for **each state** on the left panel.



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❷ The right panel visualizes the relationship between **regulatory news sentiment** and **bank profitability changes.**

❸ **It also shows the geographic distribution of banking stress across states. Interactive charts allow users to explore risk patterns across states.**

❹ **The bottom section displays the corresponding state-level data outputs.**